

Coding & Solution

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This section evaluates the quality and completeness of your implemented code and the overall solution. Your submission should demonstrate working functionality, clean code practices, and alignment with your defined requirements.

Instructions:

1. Ensure your code is well-structured, commented, and follows best practices for your chosen technology stack.
2. The solution should address the core problem statement and implement the key functional requirements.
3. Include all source files, configuration files, and setup instructions needed to run the application.
4. Attach or link to your code repository (e.g., GitHub) in the table below.

Solution Summary

Field	Details
Repository Link / URL	https://github.com/dharaniseven/Flood-prediction-project
Programming Language(s)	Python
Framework(s) Used	Flask Framework
Libraries Used	NumPy, Pandas, Scikit-learn, XGBoost, Matplotlib, Seaborn, Joblib
Key Features Implemented	1. Weather dataset analysis 2. Data preprocessing (missing values, outliers, categorical encoding, feature scaling) 3. Training multiple ML models (Decision Tree, Random Forest, KNN, XGBoost) 4. Model evaluation using accuracy score, confusion matrix, and classification report 5. Selection of best-performing model (XGBoost) 6. Flask-based flood prediction web application 7. Flood chance and no flood chance result generation
Pending / Incomplete Features	1. Real-time weather API integration 2. IoT sensor-based flood monitoring 3. Mobile application development 4. Advanced cloud deployment
Setup / Run Instructions	1. Install Python 3.8 or above 2. Install required libraries using requirements.txt 3. Load saved model files (floods.save, scaler.save) 4. Run Flask application using python app.py 5. Open browser and access http://127.0.0.1:5000/

Code Quality Checklist

S.No	Criteria	Status (Yes / No)
1	Code is modular and organized into functions / classes	Yes
2	Meaningful variable and function names are used	Yes
3	Code includes comments / documentation where necessary	Yes
4	Error handling is implemented for critical operations	Yes
5	The application runs without critical errors	Yes
6	Code is committed to a version control repository	Yes

Additional Notes / Comments:

The Rising Waters project implements a complete machine learning workflow for flood prediction, including data preprocessing, model training, evaluation, and deployment.

The system uses multiple classification algorithms and selects the best-performing model for prediction. The XGBoost model achieved 96.55% accuracy on test data and is integrated into a Flask web application.

The code structure is modular, maintainable, and designed for future improvements such as real-time weather data integration, IoT monitoring, and cloud deployment.